

## The elimination method



- 1
- a  $7$                       b  $x = -3$                       c  $y = \frac{3}{7}$
- 2
- a  $x = -\frac{16}{3}, y = 3$                       b  $x = 2, y = -9$   
c  $x = 4, y = -9$                       d  $x = 4, y = -\frac{13}{3}$   
e  $x = -4, y = \frac{23}{3}$                       f  $x = 0, y = 3$   
g  $x = -4, y = -\frac{13}{9}$                       h  $x = 2, y = \frac{11}{7}$   
i  $x = -2, y = -2$                       j  $x = -20, y = \frac{53}{3}$   
k  $x = 3, y = -5$                       l  $x = -5, y = 7$   
m  $x = -\frac{2}{3}, y = \frac{1}{3}$                       n  $x = -\frac{32}{7}, y = -3$   
o  $x = -\frac{7}{9}, y = -2$                       p  $x = 1, y = -3$   
q  $x = 5, y = -2$                       r  $x = 3, y = 1$
- 3
- a
- i  $3x + 11y = -48$  and  $-6x + 21y = -76$   
ii  $x = -\frac{4}{3}$  and  $y = -4$
- b
- i  $2x + 3y = 5$  and  $5x + 4y = 2$   
ii  $x = -2$  and  $y = 3$
- c
- i  $4x + 3y = 20$  and  $8x - 3y = 4$   
ii  $x = 2, y = 4$
- d
- i  $6x + 5y = 240$  and  $x + 3y = 66$   
ii  $x = 30, y = 12$
- 4
- a  $x = 2, y = 8$                       b  $x = 2, y = -9$                       c  $x = 8, y = 3$                       d  $x = 5, y = 5$   
e  $x = -9, y = 4$                       f  $x = 6, y = 7$                       g  $x = -20, y = 15$                       h  $x = 3, y = \frac{23}{3}$   
i  $x = -1, y = -1$                       j  $x = \frac{1}{2}, y = \frac{3}{2}$                       k  $p = -\frac{3}{2}, q = \frac{23}{10}$                       l  $x = -2, y = 3$



## Applications

5

a  $x + y = 122$

b  $x - y = 12$

c  $x = 67$

d  $y = 55$

6

a  $9x + 2y = 26$

b  $3x + 4y = 22$

c  $x = \$2$

d  $y = \$4$

7

a  $x = 1$

b  $y = 68$

c 2016

d 68%

8

a  $2x + y = 27, 5x - y = 43$

b  $x = 10, y = 7$

9

a  $x + y = 10, 6x - 5y = 5$

b  $x = 5, y = 5$

10

a  $7x + y = 64, 3x - y = 16$

b  $x = 8, y = 8$

11

a  $y = 10x, y = 7x + 18$

b Son is 6 and mother is 60 years old.